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Asymptotics of Partition Functions of Coulomb Gases

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Concentration Area: Mathematics

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"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

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For all of those who were before and are now:
Your courage, patience, daring, and caring has brought me here.
To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.
For all of those who will be after:
To promise that I can for you as it was done for me.

“João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito.”
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas de Funções Partição em Gases de Coulomb**. 2026. 42 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Este trabalho é um breve modelo para a escrita de monografias de qualificação, dissertações e teses utilizando o ambiente LaTeX, de acordo com as normas exigidas pelo Instituto de Ciências Matemáticas e de Computação (ICMC), da Universidade de São Paulo (USP). Para a confecção deste modelo foi utilizado a última versão (1.9.6) do pacote de classes *abnTeX2* que segue as normas da Associação Brasileira de Normas Técnicas. A elaboração de uma monografia, dissertação ou tese pode ser feita sobrescrevendo o conteúdo deste modelo.

Palavras-chave: Modelo, Monografia de qualificação, Dissertação, Tese, Latex.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 42 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set of such matrices with a given arbitrary distribution is called an ensemble. One can, for ensembles of interest, study statistic such as the eigenvectors and eigenvalue distributions. The symmetries of chosen set will reflect on what models such statistics can model. For example, distances of the characteristic values of a symmetric matrix with random coefficients model distances between successive levels of energies of heavy nuclei. More interestingly, there is some universality in the statistics found among classes of ensembles, suggesting the relevance of the theory in studying the universality in general modeled phenomena. Particularly important is the study of such matrices when they grow infinitely large, taking asymptotically its statistical behavior. We focus on a technique for deriving asymptotics for the so-called partition function of some interesting ensembles of random matrices.

Keywords: Random Matrix Theory, Planar Matrix Ensembles, Asymptotics.

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LIST OF SYMBOLS

He — Hermite Polynomial

$\mathbb{1}$ — Identity Function

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EQUILIBRIUM

The section on Equilibrium Measures was based on the works of Saff and Totik (2024) and Deift (1999).

4.1 Coulomb Gases

We start by describing a familiar distribution or physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of particles as a function of the system state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta n^2)$). This is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy.

Suppose a set of n particles at positions $x_1, x_2, \ldots, x_n \in \Sigma$, a *conductor* of dimension d, or some configuration $\xi_n = (x_1, x_2, \ldots, x_n)$, embedded in some space $\mathbb{R}^{dn}$.

Definition 1. *If p_n is a probability measure on $(\mathbb{R}^d)^n$, we name it **Coulomb gas** if it has the form of a Boltzmann-Gibbs probability measure, namely if*

$$\mathrm{d}\mathrm{p}_n(\xi_n) = \frac{e^{-\beta n^2 \mathcal{H}_n(\xi_n)}}{\mathcal{Z}_n^\beta} \mathrm{d}x_1 \mathrm{d}x_2 \ldots \mathrm{d}x_n, \tag{4.1}$$

where

$$\mathcal{H}_n(\xi_n) = \frac{1}{n}\sum_{i=1}^n Q(x_i) + \frac{1}{2n^2}\sum_{i \neq j}^n \mathfrak{g}(x_i - x_j) \tag{4.2}$$

is usually called the Hamiltonian, or more concretely, the configurational energy of the system. The function $Q\colon \mathbb{R}^d \cong \Sigma \to \mathbb{R}$ is the external potential.

The measure p_n models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q. If one wishes $\mathrm{d}\mathrm{p}_n$ to be a probability measure, then one must ensure that Q is such that the normalizing constant $\mathcal{Z}_n^\beta$ is finite for all n and that the measure support supp $(\mathrm{dp}_n(\chi_n))$ is compact. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. The function $\mathfrak{g}\colon \Sigma \to (-\infty, \infty]$ is the *Coulomb Interacting Kernel*, solution to the Poisson equation given by $-\nabla g(\vec{x}) = c_n \delta_0$ and is the reason in the first place why the particles may diverge.

Recall the expression for invariant ensembles in **??**, note that, rearranging the terms in Equation 4.1 and considering a logarithm interaction kernel we get

$$\mathrm{dp}_n(x_1, x_2, \ldots, x_n) = \frac{1}{\mathcal{Z}_n^\beta} \prod_{1 \le j < k \le n} |x_j - x_k|^\beta \prod_{j=1}^{n} \mathrm{e}^{-\beta n Q(x_j)} \mathrm{d}x_1 \mathrm{d}x_2 \ldots \mathrm{d}x_n \tag{4.3}$$

This is to say that for invariant ensembles, their eigenvalues behave as a Coulomb gas. This opens the way for interpretations on the distribution of eigenvalues but also to computations using simulation methods for particle dynamics.

Example 1. *As explored in Chafaï and Ferré (2018), a few classical algorithms of simulation are very successful in replicating the measures for Coulomb gas systems. Equivalently, if one is interested in sampling eigenvalues for $\beta = 2$ invariant ensembles, one can sample from its configuration equilibrium $\mathrm{p}_Q(\lambda)$ in respect to some potential V using simulation of the particles. Consider for example the non-trivial case explored in Balogh, Grava and Merzi (2016), where, taken $d = 2$ and $n = 2$ we will have $W(x) = g(x) = \log|x|$ and impose the potential*

$$Q(z) = |z|^{2a} - \mathrm{Re}\, t z^a. \tag{4.4}$$

This potential has disjoint intervals for some combinations of the parameters t and a - it presents a transition in $t_c \approx \sqrt{(1/a)}$. Analytically, it was just recently solved. However we can easily sample from this measure using the simulation as shown below. Data was taken from Pimenta and Silva (2024).

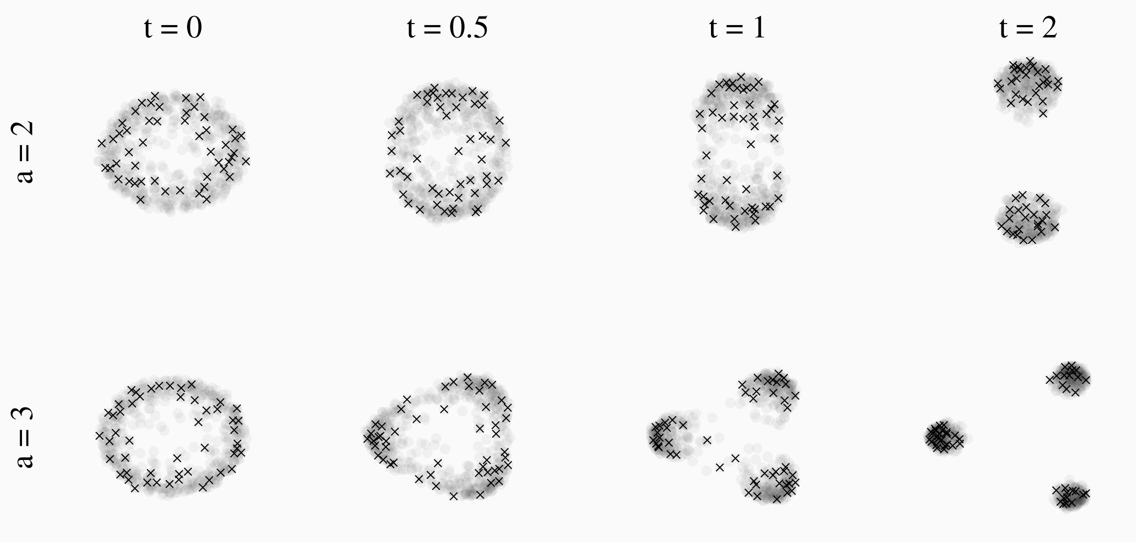

Figure 3 – Shown in the shadow is the mean configurations - in some an approximation of the limit distribution of points, and in the scatter is a random $n = 50$ sample from the distribution. We show configurations for combinations of the parameters $t = 0, 0.5, 1, 2$ from left ro right and $a = 2, 3, 5$ from top to bottom.

4.2 Equilibrium Measure

The description of the eigenvalues as a Coulomb gas and as a point process lead us to the idea of minimizing some type of energy of the system by finding equilibrium configurations $\xi_n = \{x_1, x_2, \ldots, x_n\}$. Especially interesting is the case when n tends to a large enough number as it simplifies some structures and the results.

To clarify the point, suppose we are dealing with the measure

$$\mathrm{p}_n(\xi_n) = \frac{e^{-\beta n^2 \mathcal{H}_n(\chi_n)}}{Z_{n,\beta}}, \quad \mathcal{H}_n(\chi_n) = \frac{2}{n}\sum_{i=1}^{n} Q(x_i) + \frac{1}{n^2}\sum_{i<j}^{n} \ln \frac{1}{|x_i - x_j|}. \tag{4.5}$$

For any n-configuration $\xi_n = \{x_1, x_2, \ldots, x_n\}$, let μ_x be the normalized counting measure on $\mathbb{C}$,

$$\mu_x = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \quad \text{such that} \quad \int_{\mathbb{R}} \mathrm{d}\mu_x = 1. \tag{4.6}$$

Then

$$\frac{1}{n^2}\sum_{i<j} \ln|x_i - x_j|^{-1} + \frac{2}{n}\sum_{i=1}^{n} Q(x_i) \tag{4.7}$$

may be written in the form

$$\int\int \ln|t-s|^{-1} \mathrm{d}\mu_x(t)\mathrm{d}\mu_x(s) + 2\int Q(x_i)\mathrm{d}\mu_x(s). \tag{4.8}$$

This lead us to the energy minimization problem

$$E^{(Q)} = \inf_{\mu \in \mathcal{M}^1(\mathbb{R})} \left[\int\int \ln|t-s|^{-1} \mathrm{d}\mu_x(t)\mathrm{d}\mu_x(s) + 2\int Q(s)\mathrm{d}\mu_x(s)\right] \tag{4.9}$$

$$:= \inf_{\mu \in \mathcal{M}^1(\mathbb{R})} \left[I_Q(\mu_x)\right] \tag{4.10}$$

where

$$\mathcal{M}^1(\mathbb{R}) = \left\{\mu \in \text{Borel measures on } \mathbb{R} \,\middle|\, \int \mathrm{d}\mu = 1\right\}. \tag{4.11}$$

The quantity $E^{(Q)}$ is the *equilibrium energy* of the system and $I_Q(\mu_x)$ is the *energy integral*. However, we must step backwards so that we can advance.

The classical case for such minimization problems corresponds to the cases when the conductor Σ is compact and the external field Q is zero. Under some mild conditions on the external field, however, the equilibrium measure μ_Q - the measure that minimizes such problem - has a unique solution which is a measure of compact support. For it to be compact, Q the confining potential, must increase sufficiently fast around infinity. Let $\Sigma \subseteq \mathbb{C}$ be a compact subset of the complex plane and $\mathcal{M}(\Sigma)$ be the collection of all positive Borel measure of compact support in Σ. Let $\mu \in \mathcal{M}(\Sigma)$ be such a measure in the complex plane $\mathbb{C}$.

Definition 2. *The associated* ***logarithmic potential*** *is defined by*

$$U^{\mu}(z) := \int \ln \frac{1}{|z-t|} \mathrm{d}\mu(t). \tag{4.12}$$

Also, $I(\mu)$, called the ***logarithmic energy*** *of $\mu \in \mathcal{M}(\Sigma)$ is defined*

$$I(\mu) := \int \int \ln \frac{1}{|z-t|} \mathrm{d}\mu(z)\mathrm{d}\mu(t), \tag{4.13}$$

the ***energy*** *V of Σ is*

$$V := \inf\{I(\mu) | \mu \in \mathcal{M}(\Sigma)\} \tag{4.14}$$

and the measure μ_{Σ}^{V}, if V is finite, which defines the energy is the ***equilibrium measure****.*

This logarithmic potential can be interpreted as the equivalent interaction potential of the system of n identical particles distributed with μ where the repelling force between two particles is proportional to the distance between them. There are a few useful properties one can extract about this quantity.

Proposition 1. *(SAFF; TOTIK, 2024) The logarithmic potential are super-harmonic on the whole plane, that is, they have the following properties.*

1. *$U^{\mu}(z) > -\infty$ for all z,*

2. *U^{μ} is lower semi-continuous,*

3. *$U^{\mu}(z) \geq \frac{1}{2\pi}\int_{-\pi}^{\pi} U^{\mu}(z)(x + r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta$ for all z and all $r \geq 0$.*

Proof. Firstly, property (1) follows directly by the properties of μ, especially the compact support. The property (2) can be proven noticing that

$$U^{\mu}(z) = \lim_{M\to\infty} \int \min\left(M, \ln \frac{1}{|z-t|}\right) \mathrm{d}\mu(t). \tag{4.15}$$

The functions on the right-hand side are continuous and increasing with M, such limit of an increasing sequence of functions must be lower semi-continuous. To prove (3), note that the kernel $\ln(1/|z-t|)$ is super-harmonic in $\mathbb{C}$ for each t. If some F is analytic in a domain D and $\rho > 0$, then $|F(z)|^{\rho}$ is sub-harmonic in D; furthermore, $\ln|F(z)|$ is sub-harmonic in D provided

F is not identically zero. Thus,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}\ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta \leq \ln\frac{1}{|z-t|}, \quad z,t\in\mathbb{C}. \tag{4.16}$$

Using the Fubini-Toneli theorem and the inequality above

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}U^{\mu}(z+r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta = \int\frac{1}{2\pi}\int_{-\pi}^{\pi}\ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta\mathrm{d}\mu(t), \tag{4.17}$$

$$\leq \int\ln\frac{1}{|z-t|}\mathrm{d}\mu(t) = U^{\mu}(z) \tag{4.18}$$

□

Also, for technical reasons we also state the following Lemma regarding the logarithmic energy that will become useful in a later result although we do not to prove it.

Lemma 1. *(SAFF; TOTIK, 2024) Let $\mu=\mu_1-\mu_2$ be a signed Borel measure with compact support and total mass $\mu(\mathbb{C})=0$. Suppose further that each of the positive measures μ_1 and μ_2 has finite logarithmic energy. Then the logarithmic energy of μ is nonnegative:*

$$I(\mu) := \int\int\ln\frac{1}{|z-t|}\mathrm{d}\mu(z)\mathrm{d}\mu(t), \tag{4.19}$$

and it is zero if and only if $\mu=0$.

The energy V can be finite or $+\infty$, and in the finite case there is a unique measure $\mu=\mu_\Sigma\in\mathcal{M}(\Sigma)$ for which the minimum defining energy V is reachable. Such a measure μ_Σ is called the *equilibrium distribution* or *equilibrium measure* of the compact set Σ. For such a function we have that its logarithm potential is always smaller or equal to the energy I_Σ.

Definition 3. *Define also the logarithmic capacity*

$$cap(\Sigma) := \mathrm{e}^{-V}. \tag{4.20}$$

The capacity for an arbitrary Borel set E is defined by the supreme of the capacity of all the compact subsets $K\subseteq E$. Naturally, the capacity of a subset of a Borel set of capacity zero is said to be zero. The union of zero-capacity Borel sets can also be shown to be a zero-capacity Borel set. A property is said to hold *quasi-everywhere* (q.e.) on a set E if the set of non conforming points is of capacity zero. Then, we can say that

$$U^{\mu}_{\Sigma}(z) = V, \quad \text{q.e. } z\in\Sigma. \tag{4.21}$$

We should at this point reintroduce the external potential Q. The main difference now is that we can loose the compacity restriction on the sets Σ and impose that there is no positive mass around infinity. There is a natural way to introduce this idea. Let $\Sigma\subset\mathbb{C}$ be a closed set and $w:\Sigma\longrightarrow[0,\infty)$. We call such a function a *weight function* on Σ.

Definition 4. *A weight function w is said to be* ***admissible*** *is it satisfies the following three conditions*

1. *w is upper semi-continuous;*
2. *$\Sigma_0 := \{z \in \Sigma | w(z) > 0\}$ has positive capacity;*
3. *if Σ is unbounded, then $|z|w(z) \to 0$ as $|z| \to \infty$, $z \in \Sigma$.*

Assume our weight function to have the form

$$w(z) := \mathrm{e}^{-nQ(z)}. \tag{4.22}$$

Then, for the weight to be admissible, $Q : \Sigma \to (\infty, \infty]$ must be lower semi-continuous, $Q(z) < \infty$ on a set of positive capacity and if Σ is unbounded, then

$$\lim_{|z|\to\infty, z\in\Sigma} \{Q(z) - \ln|z|\} = \infty. \tag{4.23}$$

Let $\mathcal{M}(\Sigma)$ be the collection of all positive unit Borel measures μ with support in Σ. Our definition for the weighted energy integral

$$I_w(\mu) = \int\int \ln|t-s|^{-1} \mathrm{d}\mu(t)\mathrm{d}\mu_x(s) + 2\int Q(s)\mathrm{d}\mu(s) \tag{4.24}$$

is recovered if both integrals exist and are finite. We can now move on to the matter of existence of an *equilibrium measure* μ^Q for a given external field Q given by some polynomial

$$Q(x) = t_{2m}x^{2m} + t_{2m-1}x^{2m-1} + \cdots + t_0, \quad t_{2m} > 0, V(x) \geq 0. \tag{4.25}$$

Theorem 1. *(DEIFT, 1999) Let Q be as in Equation 4.25. Then, there exists a unique probability measure $\mu = \mu^w$ such that*

$$E^w = \inf_{\mu\in\mathcal{M}^1} I^w(\mu) = I^w(\mu^w). \tag{4.26}$$

Moreover, μ^w has compact support.

4.2.1 Euler-Lagrange

Associates with the variational problem introduced on section 4.2

$$\begin{aligned} E^w &= \inf_{\mu\in\mathcal{M}^1} I^w(\mu) && (4.27)\\ &= \inf_{\mu\in\mathcal{M}^1}\left[\int\int \ln|t-s|^{-1}\mathrm{d}\mu(t)\mathrm{d}\mu(s) + 2\int Q(s)\mathrm{d}\mu(s)\right] && (4.28)\\ &= I^w(\mu^w) && (4.29)\end{aligned}$$

there are so-called *Euler-Lagrange* type variational equations. For some cases it is possible to solve these equations for μ^w.

Theorem 2. *(Variational Equations, (DEIFT, 1999)) The equilibrium measure μ^w for the given variational problem satisfies the following conditions.*
There exists a real constant l such that

1.
$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\tilde{\mu}(x) \geq l \tag{4.30}$$
for all $\tilde{\mu} \in \mathcal{M}^1(\mathbb{C})$ of compact support with $I^w(\tilde{\mu}) < \infty$.

2.
$$2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) = l \tag{4.31}$$
for μ^w-almost everywhere.

Conversely, if $\mu \in \mathcal{M}^1(\mathbb{C})$ has compact support, satisfies conditions (1) and (2) above, and $I^w(\mu) < \infty$, then $\mu = \mu^w$.

Proof. ($\Longrightarrow$) We recall that μ^w has compact support by Equation 1 and $I^w(\mu^w) = E^w < \infty$. Suppose $\mu^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu^*) < \infty$. Then, defining some convex combination

$$\mu_t = t\mu^* + (1-t)\mu^w = \mu^w + t(\mu^* - \mu^w) \in \mathcal{M}^1(\mathbb{C}), \quad 0 \leq t \leq 1, \tag{4.32}$$

we should have

$$\begin{aligned} I^w(\mu_t) = \int\int \ln|t-s|^{-1} \mathrm{d}\mu^w(t)\mathrm{d}\mu^w(s) + 2\int Q(s)\mathrm{d}\mu^w(s) \\ + 2t \int\int \ln|t-s|^{-1} \mathrm{d}\mu^w(t)\mathrm{d}(\mu^* - \mu^w)(s) \\ + 2t \int Q(s)\mathrm{d}(\mu^* - \mu^w)(s) \\ + t^2 \int\int \ln|t-s|^{-1} \mathrm{d}(\mu^* - \mu^w)\mathrm{d}(\mu^* - \mu^w)(s). \end{aligned} \tag{4.33}$$

This is all well since μ^w, μ^* have compact support and $I^w(\mu^w), I^w(\mu^*) < \infty$. As $I^w(\mu_t) \geq I^w(\mu^w)$ for $0 \leq t \leq 1$ by assumption of minimality and Equation 1, is is necessary that

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}(\mu^* - \mu^w)(x) \geq 0 \tag{4.34}$$

and therefore

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\mu^*(x) \geq l \tag{4.35}$$

where

$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) \right] \mathrm{d}\mu^w(x) := l \tag{4.36}$$

proving the first assertion in (1). Now, let

$$B = \left\{ x \,\middle|\, 2 \int \ln|x-y|^{-1} \mathrm{d}\mu^w(y) + 2Q(x) < l \right\}. \tag{4.37}$$

Then, B must be a bounded set of $\mathbb{C}$. Suppose $\mu^*(B) > 0$ and set

$$\mu_B^* = \frac{\chi_B}{\mu^*(B)}\mu^* \tag{4.38}$$

where χ_B is the characteristic function of the set B. Then $\mu_B^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I^w(\mu^*) < \infty$. inserting $\mathrm{d}\mu_B^*$ in Equation 4.35, we find

$$l \le \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x)\right]\mathrm{d}\frac{\chi_B}{\mu^*(B)}\mu^*(x) < l \tag{4.39}$$

a contradiction. Therefore,

$$\mu^*(B) = \mu^*\left(\left\{x \,\middle|\, 2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) < l\right\}\right) = 0 \tag{4.40}$$

for all measures $\mu^* \in \mathcal{M}^1(\mathbb{C})$ with support compact and $I^w(\mu^*) < \infty$. in particular, it is true for $\mu^* = \mu^w$. Since,

$$0 = \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) \tag{4.41}$$

$$= \int_{\mathbb{C}-B} \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) \tag{4.42}$$

and (2) follows.

($\Longleftarrow$) If, now, $\mu \in \mathcal{M}^1$ satisfies (1) and (2), $I^w(\mu) < \infty$, and suppose μ has compact support. Then write $\mu^w = \mu + (\mu^w - \mu)$. We have, as before

$$I^w(\mu) \ge I^w(\mu^w) \tag{4.43}$$

$$\begin{aligned} &= I^w(\mu) + \int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) - l\right]\mathrm{d}(\mu^w - \mu)(x) \\ &\qquad + \int\int \ln|t-s|^{-1}\mathrm{d}(\mu^w-\mu)(t)\mathrm{d}(\mu^w-\mu)(s). \end{aligned} \tag{4.44}$$

Using (2),

$$\int \left[2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) - l\right]\mathrm{d}\mu^w(x) - l \ge l - l = 0 \tag{4.45}$$

but also the third term is strictly positive by Equation 1 unless it is identically null. If, however, $\mu^w - \mu \ne 0$, then we showed that

$$I^w(\mu) \ge I^w(\mu^w) > I^w(\mu), \tag{4.46}$$

a contradiction. Thus, if μ satisfies $(1), (2)$, then it must be μ^w. □

If there is the knowledge that

$$\mathrm{d}\mu^w = \phi(x)\mathrm{d}x \tag{4.47}$$

for some continuous function ϕ of compact support, then

$$2\int \ln|x-y|^{-1}\mathrm{d}\mu(y) + 2Q(x) \tag{4.48}$$

is continuous and the previous theorem takes the following form

Theorem 3. *(Strong Variational Equations, (DEIFT, 1999)) Suppose that* $\mathrm{d}\mu^w = \phi(x)\mathrm{d}x$ *for some continuous function* ϕ *of compact support. Then the conditions of Equation 1 can be replaced by*

1.
$$2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) \geq l \tag{4.49}$$
for all $z \in \mathbb{C}$.

2.
$$2\int \ln|x-y|^{-1}\mathrm{d}\mu^w(y) + 2Q(x) = l \tag{4.50}$$
on $\{z : \phi(z) > 0\}$.

4.3 Recovering the Equilibrium Measure

We mentioned Equation 4.47 the possibility of knowing the density ϕ of the equilibrium measure μ^w in relation to the Lebesgue measure. We will know determine ϕ. As we did many times, assume that μ is finite and of compact support in the complex plane. For notation, let A denote the two-dimensional Lebesgue measure on $\mathbb{C}$ and Δ denote the Laplacian operator

$$\Delta F := \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2}. \tag{4.51}$$

We have the theorem below.

Theorem 4. *(SAFF; TOTIK, 2024) If in a region* R *the potential* U^μ *has continuous second partial derivatives, then* μ *is absolutely continuous in* R *with respect to two-dimensional Lebesgue measure* A*, and on* R *we have the formula*

$$\mathrm{d}\mu = -\frac{1}{2\pi}\Delta U^\mu \mathrm{d}A \tag{4.52}$$

CHAPTER

5

ASYMPTOTICS

CHAPTER

6

PLANAR ENSEMBLES

BIBLIOGRAPHY

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